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Label-Free Photonic Biosensors: Key Technologies for Precision Diagnostics

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With the rise of multidrug-resistant pathogens, emerging zoonotic viruses, and the increasing prevalence of cancer, the modernization of clinical diagnostics has become a critical priority. Label-free optical biosensors, including those based on plasmonics and silicon photonic technologies, offer promising solutions for highly sensitive molecular analysis, rapid assay times, and decentralized operation. However, important challenges must be addressed for their successful clinical implementation, including the miniaturization and integration in compact, automated devices and the direct analysis of complex human specimens, like blood. In this context, an overview of recent advancements and future perspectives in photonic biosensors are provided for clinical diagnostics. The critical limitations of these technologies are examined, and specific research objectives that need to be met to fully leverage the potential of photonic biosensors and accelerate their adoption in the clinical practice are proposed. Finally, the most innovative biosensor designs and methodologies are highlighted, along with their prospects for contributing to precision diagnosis and personalized medicine.

1. Introduction

Advances in precision medicine are transforming how diseases are managed and treated, introducing novel cell engineering techniques and nanotechnologies to increase therapeutic efficacies, and moving toward a patient-centered approach that enables a personalized treatment and individual monitoring. However, current diagnostic techniques are lagging behind with these demands. Standard clinical diagnostic methods like microbial cultures, colorimetric or fluorescent immunoassays, and polymerase chain reaction (PCR), although well established and reliable, have critical drawbacks. Cultures are limited in sensitivity and often require days to produce results, delaying treatment administration. Immunoassays and PCR are faster and highly sensitive, but they depend on expensive equipment and skilled operators, making it less accessible and unsuitable for large-population screening. These limitations are especially problematic in tackling today's healthcare challenges, including drug-resistant bacterial infections and complex conditions like cancer. To address these needs, diagnostics must evolve to deliver faster, sensitive, and more informative results. Next-generation diagnostics are envisioned as compact, automated technologies able to perform multiplexed analysis in a few minutes, providing accurate identification of the disease at early stages together with prognostic information and therapeutic guidance.

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Photonic biosensor technologies hold promising potential for the implementation as precision diagnostic systems. Photonic biosensors working in a label-free scheme (i.e., without fluorescent, radioactive, or enzymatic labels) enable real-time, highly sensitive detection, and quantification of target analytes, offering a simplified, reagent-less workflow.[1,2] These technologies operate on the principle of the evanescent field mechanism (**Figure 1**). Briefly, when light is confined or propagates along a material (e.g., a nanostructure or waveguide), a portion of the electromagnetic field extends beyond the surface into the surrounding medium. This field intensity decays exponentially with distance but is highly sensitive to changes of the refractive index near the sensor surface. Biomolecular interactions occurring within the evanescent field alter the local refractive index, which can be detected and measured by interrogating specific light properties, like intensity or wavelength. This sensing mechanism can provide excellent sensitivities, reaching femtomolar detection in a direct assay format, for a wide range of targets, including proteins, nucleic acids, small molecules, and pathogens or cells. Compared to other sensing modalities, such as electrochemical and mechanical sensors, the use of light-based transducers confers an enhanced environmental robustness, being immune to electromagnetic interferences and mechanical drifts, and a high stability and reusability. Furthermore, their potential for integration and automation makes them a promising platform for developing precision diagnostic tools that facilitate rapid point-of-care (POC) testing and enable a faster and more accurate clinical decision-making.

The most popular label-free photonic biosensor is the surface plasmon resonance (SPR) system, now regarded as a mature technology with numerous commercially available instruments. SPR biosensors are extensively employed in pharmaceutical and biomedical research laboratories due to their proven reliability and advantageous performance in characterizing kinetic and affinity properties of biomolecular interactions, which drives their widespread use in drug screening, development, or quality control

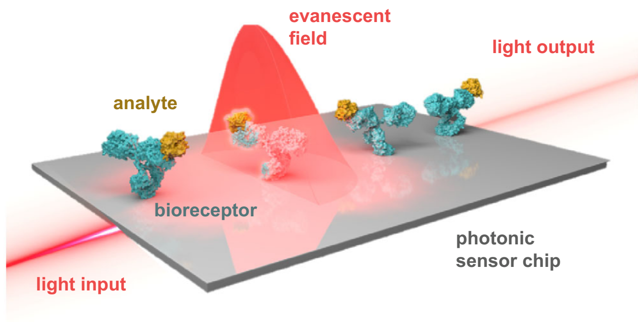

Figure 1. Illustration of a photonic biosensor working within the evanescent field principle. Biorecognition elements (e.g., antibodies, in blue) are immobilized on the sensor surface for selective capture of the target analyte (e.g., protein, in yellow). Upon light input, an electromagnetic field is generated at the interface sensor-sample with an evanescent penetration depth (i.e., decaying intensity). The biomolecular interaction alters the refractive index of the medium, changing the properties of the light output, which can be interrogated in real time and directly related to the concentration of the target analyte in the sample.

applications. Alongside SPR, nanoplasmonic sensor technologies appeared as a natural evolution, aiming to expand the applicability of label-free plasmonic sensing to clinical analysis. The incorporation of nanostructures on the sensor surface, such as nanoparticle or nanohole arrays (NHA), introduced new sensing mechanisms like the localized surface plasmon resonance (LSPR) or the extraordinary optical transmission (EOT) (**Figure 2**b). For details on the specific sensing mechanisms of these technologies, we refer to specialized reviews and scientific tutorials.[3,4] These phenomena could significantly enhance analytical sensitivity, reaching detection limits down to the single-molecule level. More importantly, these advancements contributed to improve sensors' multiplexing capabilities (i.e., parallel detection of multiple targets from a sample or parallel analysis of different samples) and to reduce the footprint of light sources, optical coupling, and detection systems, facilitating their integration into compact and

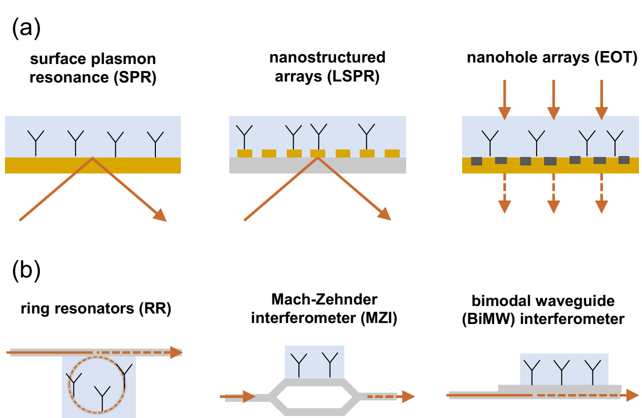

Figure 2. a) Schematics of different nanoplasmonic sensor configurations, including surface plasmon resonance (SPR) based on thin gold layer, localized surface plasmon resonance (LSPR) based on gold nanostructures, and extraordinary optical transmission (EOT) based on gold nanohole arrays and b) schematics of different silicon photonics sensor configurations, including ring resonators (RR), Mach-Zehnder interferometers (MZI), and bimodal waveguide (BiMW) interferometers.

portable devices. In research settings, nanoplasmonic biosensors have demonstrated a broad range of biomedical applications, including disease diagnosis, biomarker discovery, and therapy evaluation.[5,6] However, the inherent complexity of these biosensor technologies—requiring sophisticated fabrication processes, optoelectronic and microfluidics integration, and tailored sensor biofunctionalization, has so far hindered their technology transfer and implementation in clinical environments.

On the contrary, silicon photonics technologies emerged as direct competitors for plasmonic sensors. Biosensors based on ring resonators (RR) and interferometers, such as Mach-Zendher or bimodal waveguide (BiMW) technologies (Figure 1c), exhibit superior analytical sensitivity and potential for miniaturization, high-throughput detection, and scalable production through conventional microchip fabrication techniques. Such technologies are also based on the evanescent field sensing principle occurring in waveguides due to light total internal reflection mechanisms during propagation. In-depth explanation of silicon photonic sensing principles and designs can be found in expert technical reviews.[7–10] A notable example is the development and commercialization of multiplexed ring resonator sensors as diagnostic systems (i.e., Maverick™ Diagnostic System, Genalyte), which has received clearance from the American Food and Drug Administration (FDA) for two specific purposes: COVID-19 serology and ribonucleoprotein assay.

However, such systems still face critical limitations for their broad application in clinics, mainly related to the preparation and distribution of ready-to-use biosensor chips, and the technology integration in user-friendly, fully automated devices. The adoption of label-free photonic biosensors in the clinical practice requires the deployment of "black-box" instruments that deliver accurate and reliable information directly from a human sample (i.e., blood, serum, urine, etc.), with minimal manipulation by non-specialized operators. In this perspective, we provide a critical discussion of the main weaknesses and limitations of photonic biosensors as diagnostic systems and the latest progress in this direction. First, we analyze two key challenges in developing medical photonic biosensors: the engineering of multiplexed POC devices and the methodologies for sensor biofunctionalization. Next, we review a few recent and relevant studies that demonstrate the capabilities of photonic biosensors for clinical diagnosis, defining primary application areas and main advantages and limitations compared to traditional diagnostic techniques. Finally, we propose and highlight future application lines that could fully leverage the potential of photonic biosensors, contributing to the advancement of next-generation precision diagnostics and personalized medicine.

2. Engineering Photonics Integrated Systems

A major goal in photonics biosensor research is the development of POC diagnostic technologies, implemented as portable devices with user-friendly automated operation. The rise of mobile health technologies and the increasing demand for rapid, on-site diagnostic tools have driven significant progress in this area. The engineering of POC-integrated photonic biosensors aims, on one

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hand, to democratize healthcare, making diagnostic accessible in remote or underserved locations, and on the other hand, to accelerate the diagnostic workflow in emergency settings, to increase the efficiency of large population disease screening, and ultimately to reduce healthcare costs.

Highest-performance photonic biosensors typically rely on sophisticated and specialized instrumentation, such as microscopes, spectrometers, and powerful lasers. These technologies achieve remarkable sensitivities in the femtomolar (fM) to picomolar (pM) range, and can offer high multiplexing capabilities.[11–13] However, their complexity and reliance on costly equipment make them unsuitable for direct implementation as medical devices. To bridge this gap, photonic biosensors have also been integrated in compact benchtop instruments, employing laser diodes, photodetectors, or mini-spectrometers.[14–16] This approach generally maintains high sensitivity and certain multiplexing capabilities but in a more accessible format, enabling streamlined workflows and reduced operational costs. Such instruments could be well-suited for clinical laboratories, offering a balance between the precision of traditional systems and the practicality needed for routine testing. On-going research in this area aims to further integrate photonic biosensor technologies into miniaturized devices, replacing expensive instrumentation with low-cost, widely accessible components, such as light-emitting diodes (LEDs) and detectors based on charge-coupled devices (CCD) or complementary metal oxide sensors (CMOS) (**Figure 3**a).

An attractive approach has explored the integration of nanoplasmonic sensors with smartphones' high-quality cameras and light sources. These ubiquitous devices can provide an affordable and accessible platform for the development of portable biosensors. In particular, the latest smartphone cameras have the ability to capture high-resolution images and optical signals, even enabling spectral analysis with sensitivities comparable to commercial spectrometers. One of the first and most relevant works in the area was reported by Preechaburana et al. in 2012,[17] designing a prism-coupled SPR biosensor module that utilized the smartphone screen as light input and the front camera as detector (Figure 3b). The screen light intensity proven to be sufficient for SPR excitation while the camera enabled real-time detection. A proof-of-concept experiment was carried out for the detection of human β2-microglobulin, comparing the performance with standard commercially available SPR instruments. Another designed was proposed by Liu et al.[18] devising a fiber-optic SPR biosensor that can be easily adapted to a smartphone by directly connecting the input fiber to the LED flashlight and the output to the camera. By recording sequential images, they demonstrated real-time analysis of IgGs with a good sensitivity, also comparable to commercial SPR equipment. Other designs and engineering optimizations have been proposed in recent years to enhance signal-to-noise ratios, increase automation and miniaturization levels, and even decrease technology costs.[19] However, to our knowledge, none of these smartphone-based label-free plasmonic biosensors has been yet validated for clinical diagnostics, probably limited by the insufficient sensitivity for relevant clinical biomarkers. Additionally, the lack of standardization in smartphones, with significant differences in camera quality, lens alignment, and light sources

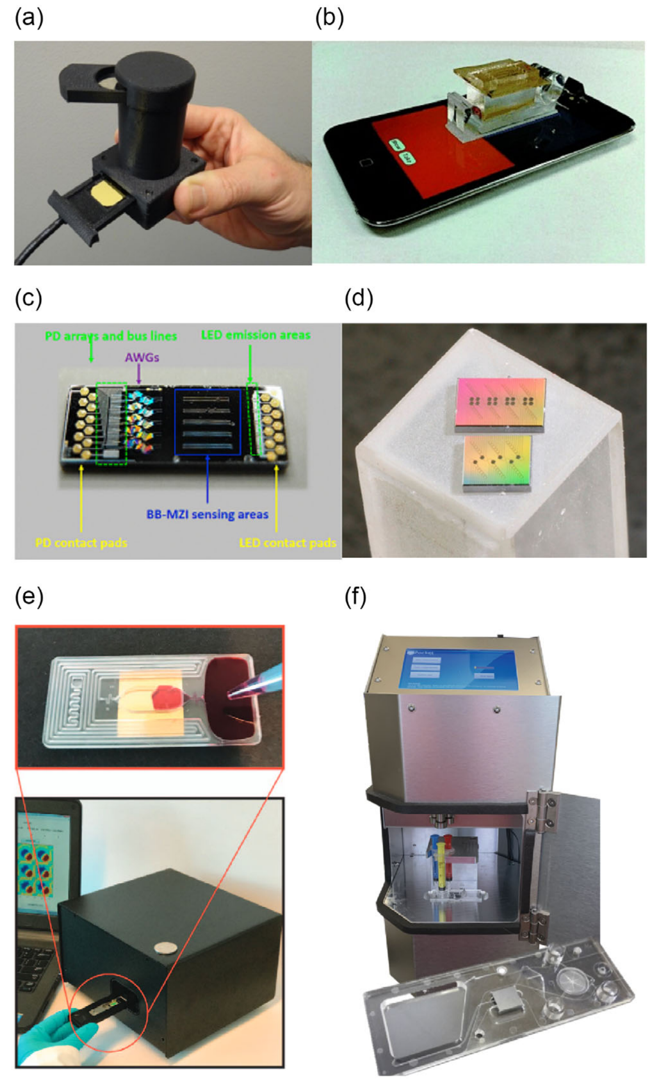

Figure 3. Examples of systems integration for label-free photonic biosensors: a) hand-held nanoplasmonic device using LEDs as source and CMOS detectors. Reprinted with permission from;[70] b) smartphone-integrated SPR sensor. Reprinted with permission from;[17] c) Monolithic integration of interferometric sensors in a photonics integrated circuit (PIC). Reprinted with permission from;[20] d) Ring resonator sensors integrated with on-chip spectrometer. Reprinted with permission from;[21] e) Microfluidics-integrated nanoplasmonic sensor (top) for operation with a compact reader (down). Reprinted with permission from;[27] and f) Standalone device for automated sensing with interferometric sensor cartridges.[32]

between smartphone models, leads to high variability in performance across devices and can introduce inconsistencies in results. This limited technology robustness may greatly hamper an eventual regulatory approval and commercialization as medical technologies.

In the area of silicon photonics, one of the key-enabling factors for the development of POC biosensors is the rising trend and progress in photonic integrated circuits (PICs). PICs are envisioned as microchip-like devices that include multiple photonic components, such as waveguides, detectors, and light sources on a single platform. A few examples of monolithic PIC sensor designs have been recently reported with proof-of-concept demonstrations. Misiakos et al. fabricated a 37 mm^2 chip integrating

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ten Mach-Zehnder interferometers, each in-coupled to a LED source and out-coupled to an array of photodiodes (Figure 3c).[20] The device was validated for the analysis of C-reactive protein (CRP) in serum, with a calculated limit of detection of 8 pM. Another interesting work was published by Bryan et al. using a series of microring resonators as on-chip integrated spectrometers for the read-out of a ring resonator sensor (Figure 3d).[21] The footprint of the device was 1×2 mm, allowing for six parallel measurements, and it was also demonstrated for highly sensitive CRP detection in serum. One of the main advantages of these PICs is the fabrication with standardized CMOS technology, which enables mass production at a relatively low cost with high reproducibility.[22,23] Moreover, they can be seamlessly integrated with electronic systems for data acquisition, processing, and wireless transmission.[24] Nonetheless, a critical parameter in the design of PIC sensors is the tailored biofunctionalization of sensor areas, requiring a high-resolution site-selective surface chemistry,[25,26] as well as ideally allowing for efficient cleaning and recycling in order to ensure cost effectivity in clinical applications. Although it can be seen as an impressive advancement in technology miniaturization, the monolithic integration of multiple optical components in a single biosensor chip would inherently increase the manufacturing expenses, greatly compromising its utility as disposable POC diagnostic tests.

To address the aforementioned challenges, the preferred approach would be developing standalone, compact, and low-cost instruments as specific readers for low-cost, exchangeable, and disposable biosensor cartridges. A few examples have shown the possibility to integrate photonic sensors in handheld systems for specific diagnostic application. For instance, Yesilkoy et al. proposed the integration of nanoplasmonic sensors in microfluidic cartridges for operation with interferometric readers at the point-of-care (Figure 3e).[27] The system prototype was tested for multiplexed detection of proteins and microRNAs,[28] and also for direct detection of E. coli in plasma samples from sepsis patients.[29] Recently, Cetin reported a portable imaging device capable of parallel interrogation of four NHA sensor areas using LED sources and a CMOS detector.[30] The nanoplasmonic biosensor was demonstrated for direct detection of sexually transmitted infection (STI) bacteria spiked in urine, achieving detection limits in the range of 10^2 colony-forming units (cfu) mL^{-1}. Although it was not validated with clinical samples either, the determined sensitivity was comparable to that previously reported for the same application using larger, laboratory-based instruments such as microscopes and spectrometers.[31] A more complete and remarkable work was developed by Ramirez-Priego et al. presenting a self-contained POC biosensor based on Mach-Zehnder interferometers for the rapid diagnosis of tuberculosis infection in urine (Figure 3f).[32] A compelling feature of the device was the integration of the sensor chip in disposable microfluidic cartridges containing injection syringes, bubble traps, and a waste reservoir to ensure user safety. The cartridges could be directly inserted in the reader, which included all optical and electronic components, flow pumping units, and a touch screen as user interface for simple and automated operation. Notably, the technology was clinically validated with 20 patient samples, demonstrating an excellent diagnostic sensitivity and specificity.

Finally, it is worth mentioning that besides systems engineering, efforts should be placed on data processing and analysis for delivering simple and understandable results as well as enhancing real-time decision-making. Here, machine learning (ML) algorithms, particularly supervised learning techniques like neural networks, could help differentiate between positive signals from background noise, thereby improving sensitivity and specificity, or could be trained to accurately interpret multidimensional data from sensors measuring different biomarkers, contributing to a higher diagnostic accuracy.[33,34] Moreover, artificial intelligence (AI)-powered models could also enable real-time diagnostic predictions by continuously monitoring biosensor outputs and comparing them to a large dataset of clinical results. This might be especially crucial in managing complex conditions like cancer or degenerative diseases.

3. Biorecognition and Surface Chemistry

The design and formation of the biorecognition layer in label-free photonic biosensors is nowadays one of the most critical hurdles for their application in clinical diagnosis. This layer typically consists of biomolecules, such as antibodies or DNA probes, which are specifically designed to bind to target analytes (e.g., proteins, pathogens, and nucleic acids). The efficiency and selectivity of this interaction directly dictates the biosensor's analytical performance, ultimately determining the diagnostic sensitivity and specificity. Besides, the antifouling capacity and stability of the biorecognition layer are key to ensure reliable results in the direct analysis of complex clinical specimens, like blood or urine, and a robust performance under different conditions. In the last decade, several studies have focused the attention on developing and comparing different sensor biofunctionalization methodologies in order to maximize detection efficiency, ensure reproducibility, and minimize nonspecific adsorptions of biological samples.[35] To that, three main components need to be optimized: sensor chemical modification, bioreceptor binding, and antifouling treatment.

Surface chemistry strategies like alkanethiol self-assembled monolayers (SAMs) and silanization have proven effective for a controlled functionalization of gold-based and silicon-based sensors, respectively. The formation of chemical matrices with tailored functionalities, including chemically reactive groups (e.g., COOH, NH_2) and relatively inert groups (e.g., OH, CH_3) allows for chemical crosslinking of the bioreceptor molecule with a determined grafting density, creating a stable and accessible biorecognition layer and reducing potential steric hindrance issues in target capturing. It is worth noting here that while surface chemistry protocols for gold sensors are well established, offering reproducibility and long-term stability, the silanization process presents significant challenges to achieve standardization.[36] The large variability in reaction conditions (solvents, temperatures, atmospheres, concentrations, etc.) as well as the surface quality differences hinder the development of universally accepted silanization methods, thereby limiting the biofunctionalization reproducibility and the broad applicability of silicon photonic biosensors.

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For bioreceptor binding, several bioconjugation techniques can be used, including the well-known carbodiimide chemistry that forms a covalent bond between COOH and terminal NH_2 groups, common homobifunctional or heterobifunctional cross-linkers targeting amine and thiol groups, or even more complex protocols, such as the click chemistry. In the case of antibodies, additional methodologies for controlling the molecule orientation have also been proposed, either based on affinity ligands such as protein A/G or calixarenes or through site-selective conjugation to biotin or DNA strands, among others.[37–39] Although ensuring an end-on orientation of antibodies is known to maximize the detection efficiency, these strategies introduce a higher degree of complexity to the biofunctionalization process and can compromise the stability and robustness of the biorecognition layer. Therefore, the selection of the immobilization strategy will depend on the nature of the biorecognition element (i.e., antibodies, DNA, proteins), availability of specific reactive groups, the assay format (i.e., direct, indirect, competitive), and the targeted performance for the diagnostic application.

Finally, as antifouling treatments, the incorporation of polyethylene glycol (PEG) chains or zwitterionic compounds in SAMs and other chemical matrices have shown significant reduction of nonspecific protein adsorptions, although not sufficient for addressing direct analysis of undiluted samples.[40] In this regard, a few works proposed the use of more complex polymers based on dopamine, acrylamide, or methacrylate to create extremely low fouling coatings.[41–43] However, the crosslinking of bioreceptor molecules to these polymer brushes is not well controlled, limiting their applicability as sensor biofunctionalization scaffolds. Further studies in this direction are crucial to facilitate translation of label-free photonic biosensors into clinical settings, where direct analysis of untreated samples like whole blood is a genuine demand. The development of innovative sensor coatings, either through the application of two-dimensional (2D) materials or new organic chemistry approaches, will be key for tackling one of the most persistent challenges in label-free biosensing technologies.

Complementing advances in surface biofunctionalization strategies, progress in biotechnology and materials sciences can also transform the landscape of biorecognition elements in photonic biosensors. Beyond traditional antibodies and DNA probes, emerging bioreceptors are offering enhanced performance, versatility, and cost efficiency. Synthetic receptors such as aptamers and molecularly imprinted polymers (MIPs) can be a relatively affordable solution for selective recognition of different target analytes, enabling an ad-hoc preparation and reproducible production at large scales.[44–46] Aptamers, which are oligonucleotide-based receptors, allow for flexible design and synthesis, incorporating specific linkers for a straightforward immobilization on the sensor surface, and they are particularly useful for detecting small molecules or peptides with limited immunogenic capacity. Similarly, MIPs provide customizable and scalable solution for creating selective recognition sites tailored to specific targets. These synthetic polymers are especially advantageous for applications requiring resilience to environmental conditions, enabling broader use of biosensors in challenging settings. However, despite these advantages, limitations persist. Aptamers often exhibit weaker binding affinity compared to monoclonal antibodies, potentially reducing their sensitivity in certain applications. MIPs, while structurally robust, can lack specificity when distinguishing between structurally similar targets and have limited capacity in capturing conformational changes or dynamic interactions. A potential, attractive alternative are nanobodies.[47] These camelid-derived antibodies are gaining traction due to their small size, high stability, and ease of production through recombinant processes. Their robust performance under diverse conditions, and the high affinity and specificity for target recognition make them ideal for biosensors targeting proteins, pathogens, and also small molecules. An example of nanobodies integration with label-free photonic sensors was proposed by Ruiz-Vega et al.[48] In their work, a bimodal waveguide interferometer was functionalized with recombinant nanobodies for the direct detection and quantification of SARS-CoV-2 viruses (**Figure 4**a). The nanobiosensor achieved sensitivities in the range of 10^2 vp mL^{-1} measuring directly in viral transport media, showing excellent reproducibility and reliability. Nonetheless, producing these nanobodies, although easier than full antibodies, still

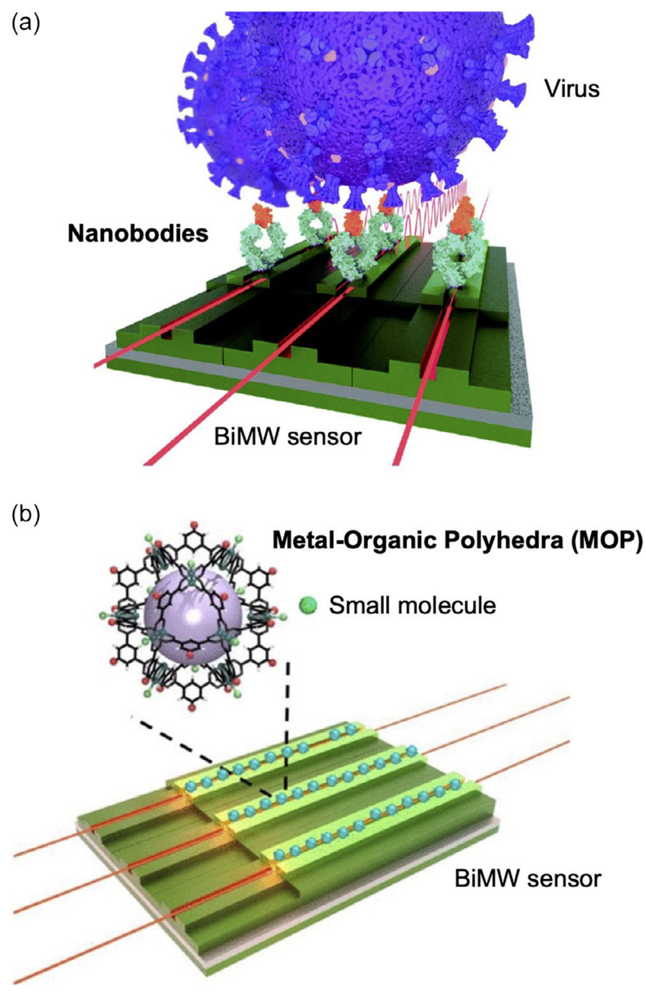

Figure 4. Examples of application of emerging biorecognition elements to label-free photonic sensors: a) Interferometric biosensor functionalized with nanobodies conjugated to antibody backbones for detection of SARS-CoV-2 viruses. Reprinted with permission.[48] and b) Interferometric biosensor functionalized with metal-organic polyhedra (MOP) for detection of small molecule contaminants. Reprinted with permission.[52]

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requires specialized host organisms like yeast or bacterial system, which may limit large-scale manufacturing as compared to synthetic receptors.

On the contrary, metal-organic frameworks (MOFs) have emerged as promising materials for biosensing.[49,50] These highly porous, crystalline, tunable structure networks provide large surface areas for specific analyte capture, being particularly interesting for small and low-abundance molecules. Although a few examples have described the application of MOFs in electrochemical sensing, their implementation in photonic technologies can be challenging due to the light interaction with the film texture and crystalline orientations, inducing scattering or diffraction effects that hamper light propagation and sensing. This hurdle could be circumvented by synthesizing miniaturized MOF particles (smaller than 30 nm), which could be assembled in optically transparent films.[51] Another interesting strategy has been recently proposed by Calvo-Lozano et al. designing metal-organic polyhedra (MOPs) with a higher solubility and the possibility for chemical functionalization on sensor surfaces (Figure 4b).[52] Although the application of these organic frameworks as biorecognition elements in photonic sensors has primarily been demonstrated for environmental monitoring, their potential for clinical diagnostics is highly promising. These materials could offer enhanced sensitivity for detecting small molecules like metabolites, peptides, or microRNAs, making them particularly advantageous for medical applications requiring high precision. Furthermore, compared to biological receptors, organic frameworks provide significantly improved robustness and stability, enabling their use in more demanding or diverse diagnostic settings. We anticipate that future research will translate these advancements into practical solutions for clinical applications, broadening their utility in real-world diagnostics.

4. Applications in Clinical Diagnostics

Given their prevalence and the significant risk for public health, the primary target applications in label-free photonic biosensors are cancer and infections. These biosensors are typically designed for direct one-step detection and quantification of established protein biomarkers (e.g., tumor markers or antibodies) in blood serum samples. Genomic biomarkers, such as microRNAs, have also been implemented through DNA hybridization assays, aiming at precise and early cancer diagnosis, prognosis and monitoring. Additionally, a few works have explored the direct identification of infectious pathogens by targeting viruses or bacteria in different body fluids, like respiratory samples or urine, as well as serological diagnosis. In **Table 1**, we provide a list of the recent and most relevant works reporting either full validation with clinical samples or significant progress toward precision diagnostics.

Accelerated by the COVID-19 pandemic, biosensors for rapid SARS-CoV-2 detection gained prominence, primarily using serological assays to identify specific antibodies. SPR biosensors have recently been validated for human and veterinary diagnostics. In one study, they showed sensitivity and specificity on par with enzyme-linked immunosorbent assay and chemiluminescence assay and outperformed lateral flow assays (LFA) when tested on 150 patient samples.[16] Another application demonstrated their effectiveness in diagnosing COVID-19 in domestic animals using 75 real samples.[53] These compact, user-friendly devices deliver results in under 15 min, ideal for primary care settings, but are limited to single-plex measurements. To address this, a multiplexed plasmonic nanohole array (NHA) biosensor was developed, enabling simultaneous differentiation of antibodies against SARS-CoV-2 antigens.[54] Validated with 14 patient samples, the device shows promise for detecting multiple infections. Additionally, silicon photonic biosensors have been reported for COVID-19 serology. Cognetti et al. introduced a disposable ring resonator cartridge with a 3 min assay time,[55] while Angelopoulou et al. developed an immersible chip with Mach-Zehnder interferometers, tested on 40 samples.[56] Though accurate, the latter requires secondary antibodies, limiting usability in point-of-care settings.

Table 1. Relevant examples of label-free photonic biosensors for diagnosis of infections and cancer.

Pathology	Sensor Technology	Target Analyte	Sample	Reference
COVID-19	SPR	Antibodies	Serum	[16,53]
	NHA			[54]
	RR			[55]
	MZI			[56]
	SPR	Viral antigen	Nasopharyngeal	[57]
	BiMW	Virus	Viral transport media	[48]
Tuberculosis	MZI	Bacterial antigen	Urine	[32]
	SPR		Sputum	[58]
	RR	Cytokines	Serum	[59]
Infectious Sepsis	NHA	Bacteria	Plasma	[29]
	NHA	Cytokines	Serum	[60]
Nosocomial infections	BiMW	Bacteria	Buffer	[61,71]
Breast Cancer	LSPR	Protein biomarkers	Serum	[13]
Ovarian Cancer	SPR	Protein biomarker	Serum	[72]
Colorectal Cancer	LSPR	Auto-antibodies	Plasma	[63]
Bladder Cancer	SPR	microRNAs	Urine	[64]
	BiMW	microRNA	Urine	[11]
Lung Cancer	BiMW	microRNA	Plasma	[65]

Photonic biosensors have also been developed for detecting viral antigens in respiratory samples to diagnose active SARS-CoV-2 infections. Chen et al. introduced a handheld SPR biosensor using light-emitting perovskite composites for Spike protein detection in nasopharyngeal samples, evaluated with four clinical samples against PCR and LFAs.[57] Interestingly, Ruiz-Vega et al. developed an interferometric BiMW biosensor for capturing and quantifying whole SARS-CoV-2 virus cells, offering accurate viral load measurements across a broad range (10^2–10^6 $TCID_{50}$ mL^{-1}), though its application to real patient samples remains untested.[48]

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In the area of bacterial infections, key applications focus on the rapid diagnosis of high-prevalence and high-risk diseases, like tuberculosis, as well as precise identification of bacterial sepsis, which is one of the major concerns in healthcare today. Tuberculosis diagnosis is challenging due to generic symptoms, latent infection states, and low sensitivity in HIV co-infected patients. Ramirez-Priego et al. developed an MZI biosensor for detecting lipoarabinomannan (LAM) in urine, showing high sensitivity and specificity in 20 clinical samples, even with HIV co-infection.[32] Other biosensors, like SPR and ring resonators, have targeted bacterial antigens (e.g., HspX) in sputum[58] or cytokine panels for diagnosing latent tuberculosis and reactivation risks.[59] On the contrary, hospital-acquired infections, mainly caused by nosocomial bacteria (e.g., *E. coli*, *P. aeruginosa*, *S. aureus*, or *K. pneumoniae*) are the main responsible for sepsis. The rapid and accurate diagnosis of this infectious sepsis is crucial for timely therapy administration and potentially saving the patient's life. Dey et al. validated a portable plasmonic NHA biosensor for detecting *E. coli* in plasma within 40 min, distinguishing infectious sepsis patients from others.[29] Another NHA device detected sepsis biomarkers (procalcitonin, C-reactive protein) with high specificity.[60] Maldonado et al. reported a bimodal waveguide interferometer for detecting *P. aeruginosa* and MRSA in under 15 min, achieving detection limits below 100 cfu mL^{-1}.[61] While not tested in biological samples, this approach shows promise for identifying sepsis agents and antibiotic resistance profiles.

Finally, label-free photonic biosensors have been extensively developed and applied for cancer diagnosis, targeting both conventional protein tumor markers and emerging biomarkers, such as autoantibodies or microRNAs. Despite significant advancements, relatively few studies achieve full clinical validation of these technologies, likely due to the challenges of obtaining well-characterized cancer patient specimens. Among the notable contributions, Yavas et al. developed a multiplexed nanoplasmonic biosensor capable of simultaneously detecting four serum protein markers associated with breast cancer.[13] Similarly, Szymanska et al. applied an SPR biosensor for detecting ovarian tumor markers in serum, demonstrating its suitability for ovarian cancer diagnosis and differentiation from endometrial cysts, with good accuracy compared to standard chemiluminescence techniques.[62] Soler et al. designed a nanoplasmonic biosensor to target two tumor-associated antibodies specific to colorectal cancer, validating the approach with clinical plasma samples.[63] This method has potential for large-scale population screening, offering the significant advantage of blood testing over invasive colonoscopies. Yueng et al. were pioneers in validating a plasmonic biosensor for direct detection of microRNA markers in clinical samples, specifically targeting miR-21 and miR-214.[64] Their work revealed clear upregulation of these markers in bladder cancer patients, with excellent accuracy when compared to PCR. Furthermore, Huertas et al. and Calvo-Lozano et al. demonstrated the application of photonic interferometric sensors for ultrasensitive detection of microRNAs in noninvasive samples. Huertas et al. achieved high specificity for bladder cancer diagnosis using urine samples,[11] while Calvo-Lozano et al. focused on plasma samples for lung cancer detection, achieving significant diagnostic specificity compared to healthy individuals.[65]

Despite these promising laboratory results, significant challenges remain for the adoption of novel diagnostic technologies in cancer care. To compete with established methods such as PCR, these technologies must offer superior performance, particularly in sensitivity, to reliably detect biomarkers at ultralow concentrations. Additionally, their multiplexing capability should allow the simultaneous identification of extensive panels of biomarkers, streamlining workflows and reducing time-to-result. Beyond these technical enhancements, there is a growing need for a comprehensive and holistic analytical approach. Advanced diagnostic platforms must go beyond merely detecting the presence of biomarkers; they should enable more informative diagnoses by integrating prognostic insights and monitoring disease progression in real time. This dual focus will be crucial for the successful integration of innovative technologies into cancer diagnostics, ultimately improving patient outcomes through personalized insights.

5. Conclusions and Future Perspective

All the studies discussed in this article serve as demonstration of the undeniable potential of photonic biosensors for clinical diagnostics. The label-free and real-time analysis together with the miniaturization and multiplexing capabilities are extremely valuable features for the development of user-friendly technologies that provide accurate results in a few minutes and decentralized operation. Although engineering and biochemistry challenges exist, the continued and rising progress in materials sciences, photonic integration, and biotechnology may have the key to overcome current hurdles and further enhance attributes and performance.

In terms of clinical applications, the high accuracy and assay simplicity of label-free photonic biosensors may provide them with unique features for addressing innovative diagnostic strategies that contribute to personalized medicine. For instance, the design of biosensors targeting specific antibiotic resistance markers in bacteria could drastically reduce the sample-to-therapy times in emergency settings, enabling a straightforward administration of the most appropriate antimicrobial treatment, and contributing to decrease the use of broad-spectrum antibiotics. Also, compact and user-friendly systems could be installed in primary care centers, pharmacies, or at the patient's bedside for individual therapeutic drug monitoring (TDM), allowing for adjusting the dosage to each specific patient, enhancing the efficacy and minimizing side effects. Finally, photonic sensor technologies have potential for aiding in the discovery of novel diagnostic tools or evaluation of advanced therapies. The application of these biosensors in complex biomedical studies, such as epigenetic alterations (e.g., DNA methylation, alternative splicing)[14,66,67] or for the analysis of cell functionality,[68,69] could help in the implementation of high-precision early tumor markers and accelerate the widespread incorporation of cell-based immunotherapies, for example.

In summary, label-free photonic biosensors can become a key technology for next-generation precision diagnostics. However, realizing this potential will necessitate interdisciplinary

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collaboration, particularly with clinicians, to address specific diagnostic needs and develop fully functional technologies with strong prospects for regulatory approval, commercialization, and adoption in the clinical practice.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: label-free analysis · optical biosensor · plasmonics · point-of-care diagnostics · silicon photonics

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